Why Do Firms Remain Informal? Empirical Evidence from Indonesia (ISES 2023 Data Analysis)

Kornelius Hasiholan Pardosi

Faculty of Economics and Business, University of Indonesia

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ABSTRACT

Despite sustained government efforts to promote formalization, a substantial share of Indonesia's micro and small enterprises (MSEs) continues to operate informally. This study examines the factors shaping firms' decisions not to register their businesses, using data from the 2023 Indonesia Informal Sector Enterprise Survey (ISES). Reasons for informality are classified into three categories: regulatory burden, information gaps, and voluntary choice. Employing a multinomial logit model, the results show that education plays a central role in shaping these perceptions. Higher levels of education significantly increase the likelihood that entrepreneurs identify regulatory burden as the primary constraint, while less-educated entrepreneurs are more likely to remain informal by choice. These findings suggest that informality among more educated entrepreneurs reflects institutional constraints rather than voluntary behavior. The results highlight the need for policy interventions that streamline licensing procedures, reduce compliance costs, and improve the accessibility of formalization systems.

1 INTRODUCTION

The informal sector plays a crucial role in Indonesia's economy as a primary labor absorber, yet it remains trapped in low productivity due to structural barriers hindering formalization (Hapsari et al., 2023). Despite government efforts to digitize business permits through the Online Single Submission (OSS) system, mass formalization of micro and small enterprises remains elusive. This highlights that formalization barriers are multidimensional; the effectiveness of formalization policies depends heavily on the state's ability to reduce the bureaucratic burden, alleviate high compliance costs, and address the lack of understanding regarding legal benefits experienced by micro-enterprises (Ablaza et al., 2023).

Consequently, the informal sector is far from homogeneous, consisting of entrepreneurs facing institutional exclusion and those who voluntarily exit the formal system based on cost-benefit calculations (Perry et al., 2007). Because these decisions are profoundly shaped by local public service quality and human capital, especially education (Rothenberg et al., 2016), this study aims to empirically examine these determinants using data from the 2023 Indonesia Informal Sector Enterprise Survey (ISES). Employing a Multinomial Logit regression model, this research explores three main categories of informality reasons: regulatory burden, limited information, and voluntary choice. By specifically focusing on how educational attainment

influences the perception of these barriers, this study seeks to provide data-driven insights for more precise formalization policies in Indonesia.

2 LITERATURE REVIEW

2.1 Informal Sector in the Context of Indonesian Economy

The informal sector is a significant pillar of the Indonesian economy, yet it is often trapped in a cycle of low productivity. Hapsari et al. (2023) explain that the informal economy in Indonesia encompasses a vast scale, both in terms of contribution to GDP and employment, with over half of the workforce engaged in informal activities. However, they highlight the significant policy challenges of integrating these businesses into the formal sector. This is echoed by Ablaza et al. (2023), who note that despite the growing digitalization and government efforts through the Online Single Submission (OSS) system, the transition to formalization remains slow due to structural barriers inherent in Indonesia's highly diversified informal sector.

2.2 Regulatory Barriers and Compliance Costs

One of the main theories in informality is the view that entrepreneurs feel the costs of becoming formal (such as taxes and licensing fees) far outweigh the benefits. Perry et al. (2007) introduced the concept of exit, suggesting that many entrepreneurs consciously choose to remain informal not because of exclusion, but rather because of a rational cost-benefit calculation. In the context of Indonesia, Rothenberg et al. (2016) identified that the barriers to the formalization of Micro and Small Enterprises (MSEs) are heavily influenced by perceptions of complex bureaucracy and high regulatory costs (regulatory burden). This suggests that many micro-entrepreneurs find better operational efficiency outside the formal system due to the prohibitive costs of compliance.

2.3 Access to Information and Local Challenges

Lack of knowledge regarding registration procedures and the benefits of becoming formal (information gap) is another significant inhibiting factor. According to the comprehensive review by Ablaza et al. (2023), opportunities for formalization in Indonesia are often hampered by a lack of effective information dissemination at the community level. Differences in institutional capacity and the quality of public services across regions lead to significant variations in how easily entrepreneurs can access registration procedures, making location a crucial determinant of informality.

2.4 The Role of Individual Characteristics and Human Capital

The decision to remain informal is profoundly influenced by the sociodemographic background and human capital of business owners. Rothenberg et al. (2016) emphasize that informal firms in Indonesia are predominantly managed by individuals with lower levels of educational attainment. Entrepreneurs with limited education tend to have restricted access to administrative information and are more likely to operate informally. Consequently, improving the educational attainment and administrative literacy of business owners is recognized as a fundamental step toward overcoming the behavioral barriers to formalization.

2.5 Application of Multinomial Logit Model in Choice Analysis

In analyzing choices that are qualitative or categorical in nature (such as the distinct reasons for remaining informal), the Multinomial Logit model is an econometrically sound method. Based on standard econometric theory by Cameron and Trivedi (2005), this model allows researchers to estimate the probabilities across multiple, unordered categorical choices. This method is highly effective for capturing how marginal changes in

independent variables (such as education or business scale) affect the absolute probability of entrepreneurs choosing one of the main reasons: regulatory burden, lack of information, or voluntary choice.

3 METHODS

3.1 Research Data and Samples

This study utilizes cross-sectional secondary data from the 2023 Indonesia Informal Sector Enterprise Survey (ISES), compiled by the World Bank. This survey comprehensively captures the characteristics of micro and small enterprises (MSEs) operating outside the official licensing system in various cities across Indonesia. To ensure the reliability and theoretical consistency of the Multinomial Logit estimation, a rigorous sample restriction was applied to the initial dataset. Specifically, observations indicating "registration not applicable" as their primary status (R3 = 8, $N = 59$) were excluded from the analysis. This exclusion is theoretically justified because this response reflects a distinct perception of business applicability rather than an explicit structural or behavioral reason for informality. Furthermore, observations classified under R3 = 5 ($N = 3$) were also removed due to an insufficient sample size to achieve reliable econometric estimation. Consequently, the final analytical sample is restricted to complete case observations that fall strictly into the three main categories of interest.

3.2 Variable Measurement

The dependent variable in this study is the perception of business actors regarding the main reasons for not registering their business, which is classified into three nominal (unordered) categories: (0) Regulatory Burden (used as the base/reference category), (1) Information Gap regarding how to register, and (2) Voluntary choice because it feels there is no benefit.

Table 1: Definition and Operationalization of Research Variables

Variable Name	Data Type	Variable Role
The Reasons for Not Registering (0=Regulatory Burden, 1=Information Gap, 2=Voluntary)	Categorical (Nominal)	Dependent
Education Level of Business Owners (1-7 scale)	Categorical (Ordinal)	Independent (Main)
Log Number of Workers	Continuous	Independent (Main)
Log Monthly Sales	Continuous	Independent (Main)
Bank Account Ownership (1=Yes, 0=No)	Dummy (Binary)	Independent (Main)
Age of Main Owner	Continuous	Control
Female Owner (1=Female, 0=Male)	Dummy (Binary)	Control
Business Sector	Dummy (Binary)	Control
City Fixed Effects	Dummy (Binary)	Control

The main independent variables comprise three constructs. First, human capital is proxied by the owner's education level. Second, business scale is measured through log-transformed values of the number of workers (log workers) and average monthly sales (log sales). Third, financial inclusion is operationalized as a dummy variable for bank account ownership. In order to minimize omitted variable bias, this model also includes a series of control variables, namely the age and gender of the owner, business sector characteristics, and city-level fixed effects.

3.3 Model Specification: Multinomial Logit Regression

Considering that the dependent variable is qualitative (categorical) and does not have an absolute hierarchical order, OLS estimation is thus inappropriate. This study therefore applies a Multinomial Logit (MNL) regression model. This study follows the discrete choice analysis standard (Cameron & Trivedi, 2005), which allows for the estimation of the probability that a business falls into the constraint category of Information Gap or Voluntary relative to the baseline category of Regulatory Burden.

The analysis proceeds through three nested model specifications. Model 1 serves as the baseline regression focusing on the core human capital and business characteristics. Model 2 introduces additional control variables, while Model 3 further incorporates city fixed effects. Model 3 is determined as the main specification because it is able to control for the heterogeneity of policies, infrastructure, and public service efficiency in each observed city.

3.4 Average Marginal Effects (AME)

The main weakness of the Multinomial Logit regression coefficient is that its value is in the form of a log-odds estimate, which cannot be directly interpreted as probability changes (Cameron & Trivedi, 2005). To overcome this, the analysis stage is deepened by calculating the Average Marginal Effects (AME) of the best model specification. The AME calculation aims to quantify the direct impact as a percentage change in the absolute probability of a specific informality reason occurring due to a one-unit change in the independent variable (for example, an increase in one level of education), assuming the other variables remain constant (ceteris paribus).

4 RESULTS

4.1 Descriptive Characteristics of Informal Businesses in Indonesia

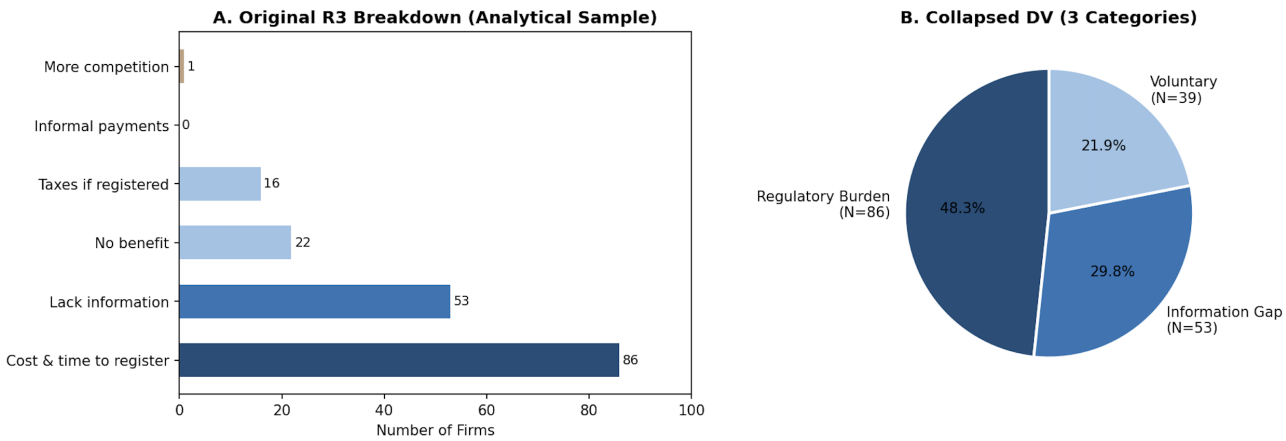

Figure 1: Distribution of Reasons for Non-Registration
Source: ISES 2023 data, processed by researchers.

Based on an analytical sample of 178 business units extracted from ISES 2023 data, the first finding highlights the proportion of primary reasons businesses fail to register their legal permits. The distribution visualization in Figure 1 shows that the regulatory burden dominated as the main obstacle, chosen by almost half of the total sample (86 business units). The remainder was attributed to a lack of information (Information Gap) comprising 53 cases, and voluntary refusal to register (Voluntary) accounting for 39 busi-

nesses. The high level of complaints about this regulatory burden provides an early indication that for the majority of micro and small businesses in Indonesia, the current licensing ecosystem is still perceived as too expensive, time-consuming, or too complicated to access.

Table 2: Descriptive Statistics of Business Characteristics Based on Informality Reasons

Variables	**Full Sample**	**Regulatory Burden**	**Information Gap**	**Voluntary**
Education (1-7 scale)	4.23 (1.10)	4.38 (1.13)	4.19 (0.92)	3.95 (1.21)
Workers (persons)	1.69 (1.10)	1.64 (0.98)	1.60 (0.84)	1.90 (1.55)
Monthly Sales (IDR)	8,944,662.92	8,683,720.93	10,503,773.58	7,401,282.05
	(7,943,784.63)	(8,179,858.48)	(8,751,394.66)	(5,785,400.04)
Owner Age (years)	45.29 (9.67)	45.50 (9.31)	44.47 (9.51)	45.95 (10.79)
Has Bank Account (%)	30.3%	25.6%	30.2%	41.0%
Female Owner (%)	42.1%	37.2%	47.2%	46.2%

Source: ISES 2023 data, processed by researchers.

Furthermore, an exploration of business characteristics across each group, as detailed in Table 2, reveals highly contrasting profiles. Entrepreneurs constrained by the Regulatory Burden exhibit the highest average educational attainment, whereas the Voluntary group comprises entrepreneurs with the lowest education levels and the smallest operational scale. Interestingly, businesses facing an Information Gap recorded the highest average sales, indicating substantial economic potential that remains outside the formal sector due to inadequate socialization of registration procedures. Conversely, a paradoxical finding emerged regarding financial inclusion, the Voluntary group exhibited the highest bank account ownership rate (41.0%) across all cohorts. This suggests that connectivity to basic banking services alone is insufficient to incentivize the formalization of business entities.

4.2 Determinants of Reasons for Informality: Multinomial Logit Regression Results

Table 3: Multinomial Logit Regression Estimation Results on Reasons for Informality (Basic Category: Regulatory Burden)

Model / Variable	Education	Log(Worker)	Log(Sales)	Has Bank Account
Model 1				
Info Gap ($j = 1$)	-0.205 (0.171)	-0.097 (0.399)	0.223 (0.200)	0.174 (0.401)
Voluntary ($j = 2$)	-0.416** (0.200)	0.523 (0.427)	-0.264 (0.198)	0.863* (0.444)
Model 2				
Info Gap ($j = 1$)	-0.201 (0.177)	-0.113 (0.404)	0.260 (0.204)	0.153 (0.405)
Voluntary ($j = 2$)	-0.382* (0.204)	0.558 (0.435)	-0.275 (0.204)	0.883** (0.448)
Model 3 (Main)				
Info Gap ($j = 1$)	-0.347* (0.199)	-0.492 (0.466)	0.179 (0.224)	0.199 (0.434)
Voluntary ($j = 2$)	-0.509** (0.231)	0.339 (0.503)	-0.227 (0.224)	0.766 (0.475)

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Source: ISES 2023 data, processed by researchers.

To empirically test these determinants, a Multinomial Logit estimation was conducted across three model specifications (Table 3), using Regulatory Burden as the base outcome. Model 3, which incorporates city fixed effects, was selected as the main specification because it provides the most robust estimation by accounting for unobserved regional heterogeneity. This approach aligns with the findings of Rothenberg et al. (2016), who emphasized that variations in infrastructure and bureaucratic efficiency at the city level significantly influence business actors' perceptions of the formalization process.

The regression results show that business scale, as proxied by the log number of employees and the log number of sales, does not have a statistically significant effect. This challenges the traditional assumption that informality is solely driven by the inability of small businesses to achieve economies of scale. Instead, human capital, namely education level, plays a crucial predictive role. The education variable has a negative and significant coefficient on the probability of choosing Information Gap (-0.347) and Voluntary (-0.509). This finding aligns with the research of Rothenberg et al. (2016), which suggests that as entrepreneur literacy and educational attainment increase, business owners are no longer hampered by a lack of information or a reluctance to participate, but are instead more keenly aware of the structural barriers, such as the expensive and complicated licensing process (regulatory burden).

4.3 Probability Analysis: Average Marginal Effects (AME)

To quantify the absolute magnitude of these impacts, the calculation of Average Marginal Effects (AME) from Model 3 yields more precise probability insights.

Table 4: Average Marginal Effects (AME) - Model 3

Variable	Education	Log(Workers)	Log(Sales)	Has Bank Account
Reg. Burden ($j = 0$)	0.0857** (0.0343)	0.0323 (0.0826)	-0.0030 (0.0383)	-0.0889 (0.0774)
Info Gap ($j = 1$)	-0.0325 (0.0339)	-0.1138 (0.0790)	0.0480 (0.0385)	-0.0115 (0.0737)
Voluntary ($j = 2$)	-0.0531* (0.0308)	0.0815 (0.0676)	-0.0450 (0.0303)	0.1004 (0.0630)

Note: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$
Source: ISES 2023 data, processed by researchers.

The results show that each increase in an entrepreneur's education level increases the probability of the business being hampered by Regulatory Burden by 8.57 percentage points (statistically significant). Simultaneously, a one-unit increase in educational attainment decreases the probability of an entrepreneur choosing voluntary informality by 5.31 percentage points.

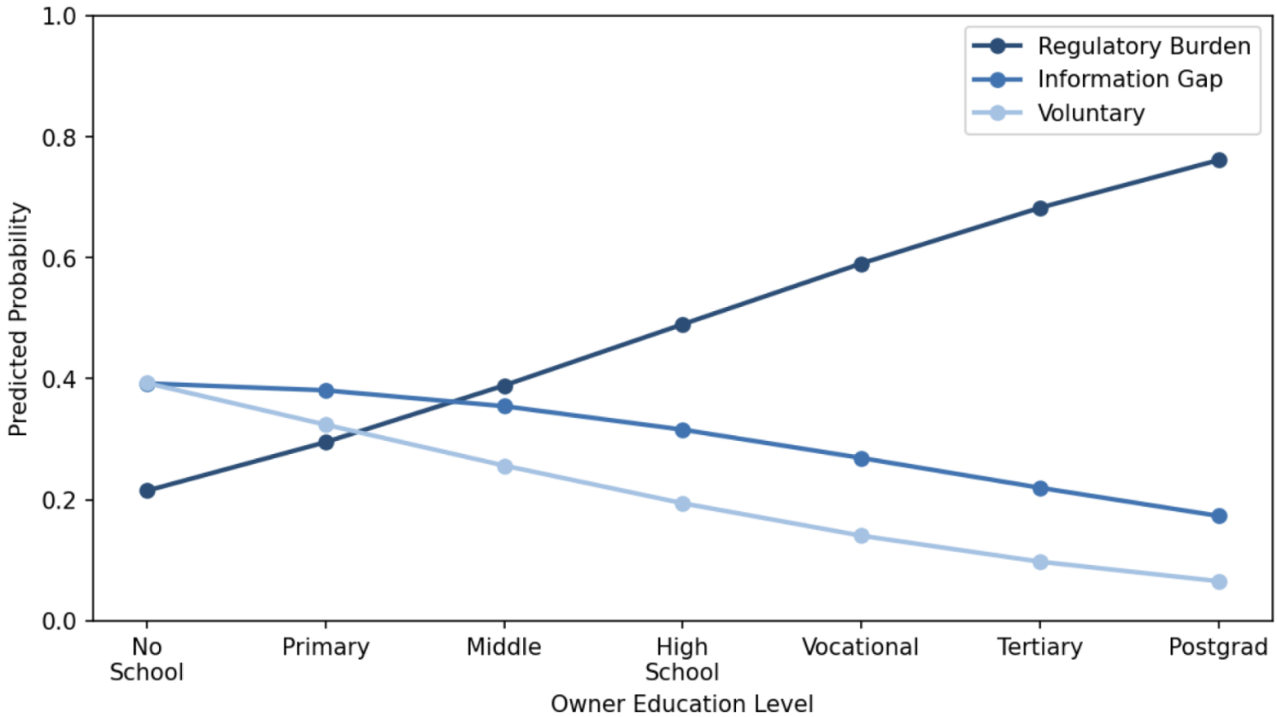

Figure 2: Prediction of the Probability of Choosing Reasons for Informality Based on the Education Level of Business Owners
Source: ISES 2023 data, processed by researchers.

This probability trajectory is clearly visualized on the prediction probability curve (Figure 2), where the line for Regulatory Burden rises steeply as the level of education increases, while the lines for Voluntary and Information Gap consistently trend downward. The implications of this finding are significant, education has successfully shifted the entrepreneur paradigm from unwilling and unaware to wanting to register, but it's too complicated. Consistent with the framework by Perry et al. (2007), this demonstrates that for educated entrepreneurs, informality is no longer a voluntary choice (exit) but rather a forced exclusion due to the failure of institutions to provide efficient and inexpensive

bureaucratic services, a challenge also highlighted in the recent findings of Hapsari et al. (2023).

5 Model Limitations and Robustness

Despite the robust theoretical alignment of the findings, several methodological limitations of this study must be acknowledged. Primarily, the most comprehensive specification (Model 3) incorporates 26 parameters, driven largely by the inclusion of city fixed effects, against a relatively modest sample size of 178 observations. This high parameter-to-observation ratio reduces the degrees of freedom and introduces a potential risk of model overfitting. While this limits the generalizability of the exact point estimates, the inclusion of regional controls was theoretically imperative to account for unobserved heterogeneity in local bureaucratic efficiency.

To mitigate this limitation, this study presented stepwise estimations (Models 1 and 2), which demonstrated that the direction and significance of the core human capital variables remained remarkably stable across different specifications. Nevertheless, future research should aim to validate these trajectories utilizing a larger, nationally representative dataset to improve statistical power.

6 CONCLUSION

This research demonstrates that the informal sector in Indonesia is not a homogeneous entity but is instead shaped by motivations and constraints that are highly dependent on the human capital of its managers. The Multinomial Logit estimation results show that education level is the most crucial determinant in shifting the direction of formalization decisions. Increased literacy and education significantly reduce the probability of entrepreneurs remaining informal voluntarily; however, they simultaneously triggers a sharp increase in the probability of concerns regarding regulatory burdens. This finding confirms that for educated micro and small business actors, informality is not a result of deliberate reluctance (exit) but rather a form of institutional exclusion. These individuals are fundamentally willing to legalize their businesses but are hampered by a licensing bureaucracy that is still perceived as prohibitively complicated and expensive.

Consequently, policy interventions must evolve beyond basic socialization and information dissemination campaigns. To effectively capture this demographic of entrepreneurs who are ready to formalize, the government must prioritize structural reforms that genuinely streamline the licensing process and reduce actual compliance costs. Addressing local bureaucratic inefficiencies, such as optimizing the execution of the Online Single Submission (OSS) system at the regional level, is critical to transforming institutional exclusion into formal economic integration.

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